

33. Assessing simple Markov Chain Models for Price Changes in a LOB

The idea here is to use a very basic Markov Chain model to capture the essential dynamics of high-frequency price changes in a LOB. Such changes are known to be serially correlated. Can a simple MC model mimic this behaviour? We will provide a large dataset of bid and ask price histories from the crypto exchange Hyperliquid, and other histories may also be available. The project is a little more speculative than the other two and would suit a student with good data-handling and analysis skills and some knowledge of time series analysis, tests for goodness of fit and serial independence, etc.

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34. Deep Learning Models of Loan Portfolios and Asset-backed Securities

The commercial and residential loan market is one of the largest asset classes with over \$20 trillion in value. Banks and financial institutions hold large loan portfolios and investment funds trade asset-backed securities (ABS) with underlying pools of small business, mortgage, credit card, commercial, and student loans. The cashflows from these loan portfolios are a function of the number of loans which are current, delinquent, prepaid, or defaulted. Each loan has a high-dimensional set of data features, which can be used to model their likelihood of becoming delinquent or defaulting. In this dissertation, a deep learning model will be developed and trained to predict loan delinquency and prepayment. The model will be benchmarked against simpler statistical models (such as logistic regression). A large dataset of 30+ million loans with billions of monthly data samples will be used to train and evaluate models.

35. Deep Learning Methods for solving High-dimensional Partial Differential Equations

High-dimensional partial differential equations (PDEs) are computationally challenging for traditional numerical methods to solve. For example, the number of grid points for finite difference methods will grow exponentially with respect to the number of dimensions, quickly becoming intractable even for high-performance computing (HPC). Recently, mesh-free deep learning methods have been proposed to solve high-dimensional PDEs. These methods will train a deep neural network to satisfy the PDE at randomly sampled points using stochastic gradient descent. Since a grid is never formed, deep learning methods avoid some of the challenges with higher-dimensional PDEs. The objective function is highly non-convex, since it involves a PDE operator applied to a neural network. This project will explore using second-order optimization methods, instead of stochastic gradient descent, to train deep neural networks to solve high-dimensional PDEs. The second-order optimization methods will be implemented using HPC/GPUs to accelerate the training.